Compact Calibrated Prototypical Networks for Cross-Domain Few-Shot Plant Disease Classification

Abstract*—Deep neural networks for plant disease classification exhibit nearly perfect accuracy in lab-controlled settings but suffer severe performance degradation in field conditions. In addition, out-of-distribution field images are assigned uncalibrated, overconfident probabilities by conventional softmax classifiers. In this paper, we propose a compact (4.26M parameters) few-shot meta-learning network utilizing multi-scale feature fusion, learned temperature-scaled metric inference, and query-level regularization. Tested using a stringent 5-way 5-shot episodic evaluation with six independent seeds (three-seed subset selected for matched-pair comparison over 1,800 episodes), our model outperforms the simple baseline Prototypical Network architecture by 8.11 percentage points (seed-level paired t-test, $p < 0.01$; episode-level effect size $d = 0.66$) on the PlantDoc field challenge dataset. Under outdoor domain shift, Expected Calibration Error is reduced from 29.30% in-domain to 3.20% in-field. A supervised nearest-class-mean baseline achieves comparable cross-domain accuracy but yields a field ECE of 12.56% versus our 3.20%, positioning our full pipeline as a calibrated framework that exposes critical failure modes in confidence-based abstention. Through 24 runs of architectural screening with a grid search, we show that dual attention blocks reduce domain generalization ability by 4.68 percentage points ($p \approx 0.033$, $df = 2$). We discover a key failure point where confidence-based selective prediction fails non-monotonically for novel field categories. Together, these findings show that compact, calibration-aware meta-learning can achieve reliable lab-to-field transfer, and identify the failures—overconfident abstention on novel healthy crops under field shift—that must be resolved before safe deployment of AI-assisted crop monitoring.

***Index Terms*—few-shot learning, cross-domain generalization, prototypical networks, calibration, plant pathology, selective prediction**

I. Introduction

Automatic recognition of plant leaf diseases is essential for effective agricultural intervention and food security [1]. Although recent convolutional neural networks achieve >98% classification accuracies on carefully curated datasets like PlantVillage [2], experiments show that these models fail on in-the-wild field data like PlantDoc [3] losing 40–50 percentage points of accuracy. The reason behind this performance drop can be attributed to three factors:

1) *Visual domain shift:* Domain shift between clean isolated images of leaves in a laboratory setting vs complex field imagery with dirt clutter, partial occlusion, dynamic light conditions and overlap of foliage.
2) *Taxonomic shift:* Class shift due to encountering disease strains or asymptomatic crops outside of the training distribution in real-world settings.
3) *Miscalibration:* Model miscalibration, i.e., the standard softmax classifier is too confident on out-of-distribution errors [4], preventing downstream tasks from safe abstention.

Few-Shot Learning (FSL) enables the classification of novel classes using limited reference examples. In agricultural deployment, this necessitates a **source-support / target-query** paradigm—using clean laboratory references to classify in-the-wild field imagery—yet prior cross-domain plant FSL work does not analyze calibration or abstention under joint domain and class shift, nor benchmarks whether episodic meta-learning provides any benefit over standard supervised pretraining. In addition, improvements via attention mechanisms have not been verified on models using squeeze-and-excitation (SE) blocks such as EfficientNet-B0.

In our work, we propose a calibrated and compact few-shot meta-learning approach for cross-domain field deployment. The main contributions of this paper are:

- **A compact, computationally efficient architecture** that fuses multi-scale features into a temperature-scaled hyperspherical embedding, enabling fast deterministic inference.
- **A rigorous multi-seed paired statistical evaluation** demonstrating statistically significant cross-domain transfer superiority over the vanilla episodic baseline.
- **A three-paradigm benchmark** (episodic, supervised, vanilla) revealing that while standard pretraining matches cross-domain accuracy, our full pipeline provides uniquely calibrated predictive probabilities.
- **Principled negative results** from a systematic architectural screening, demonstrating that dual attention mechanisms degrade cross-domain transfer and Monte Carlo Dropout incurs prohibitive latency without accuracy gains.
- **Identification of a critical selective prediction failure mode** where confidence-based abstention behaves non-monotonically on novel field classes.

The rest of this paper is structured as follows. Section II provides literature review. Section III outlines our architecture. Section IV details the experimental setup. Section V reports benchmarking results. Section VI provides ablations and negative results. Section VII analyzes calibration and failure modes. Section VIII identifies limitations, and Section IX concludes.

II. Related Work

A. Cross-Domain and Few-Shot Plant Disease Recognition

Metric-based FSL methods like Prototypical Networks [5] and Matching Networks [6] train transferable embeddings using only a small number of support examples, whereas optimization-based methods like MAML [7] optimize model parameters using inner loop gradient updates. Much plant disease literature uses image-level splits (measuring memorization, not generalization [8], [9]), which are incomparable to class-disjoint evaluation. In agriculture, AgriFewNet [10] uses ResNet-18 and dual attention with MAML to report 97.1% at 10-shot on PlantVillage, whereas compact ResNet-18 Prototypical networks [11] demonstrate strong few-shot classification performance on PlantVillage without attention mechanisms. Yang et al. [12] and SWE-MAML [13] study cross-domain adaptation in laboratory and field settings, with SWE-MAML achieving 75.71% at 30-shot for potato disease transfer. PlantCaFo [14] performs few-shot cross-domain transfer using CLIP/DINO foundation model features, reporting 67.4% on a multi-crop novel-class split, while ViT+MoE [15] reports 68% PV→PlantDoc accuracy under an all-classes-seen setting, with entropy and orthogonal regularization that resembles our orthogonality prototype term. Billion-parameter foundation models [16] give good transfer results but do not respect the computational constraints of resource-limited agricultural deployment. Though these studies use larger backbone models compared to our 4.26M-parameter model, and report accuracy under different protocols, none study calibration or selective prediction; we instead isolate domain shift from class novelty and analyze abstention under a strict class-disjoint protocol (precluding direct accuracy comparisons with all-classes-seen works).

B. Calibration and Selective Prediction in Few-Shot Learning

Contemporary neural networks generate miscalibrated confidence estimates [4]. The Expected Calibration Error (ECE) [17] is the standard diagnostic metric for quantifying this miscalibration. Selective classification [18] makes use of confidence scores to perform selective prediction, which is possible on the assumption that risk decreases monotonically with coverage. Related to our work, Fatema and Maradithaya [19] employ Prototypical Networks along with Monte Carlo Dropout (MCD) for few-shot uncertainty evaluation, yet only in the context of in-domain few-shot image classification. We consider uncertainty performance under the lab-to-field shift, namely, we identify a case of calibration inversion (ECE 29.30% in-domain → 3.20% in the field), a domain-shift-specific failure of selective prediction (increase in risk is not monotonic with abstention), and an argument based on deployment cost against the use of MCD ($10.1\times$ latency, zero accuracy gain); none of which [19] addresses.

C. Attention Mechanisms in Compact Plant Models

Dual attention (channel and spatial) layers like CBAM [20] have been extensively used to improve discriminative feature generation. AgriFewNet [10] uses dual attention in ResNet-18 backbone and finds an improvement in few-shot learning accuracy. But attention helps in a backbone and integration-specific manner: EfficientNet-B0 [21] uses SE layers at all inverted residual layers. In this work, we hypothesize that the additional sigmoid gate re-calibrates the already calibrated channels through the SE layer (see Section VI for the ablation result).

III. Methodology

A. Backbone and Multi-Scale Feature Fusion

Fine-resolution plant pathology diagnosis requires localization of not only the small lesions (spots, pustules) but also the entire leaf morphology. The proposed system uses ImageNet pre-trained EfficientNet-B0 architecture [21] with differential learning rates ($\text{lr}_{\text{backbone}} = 10^{-5}$, $\text{lr}_{\text{head}} = 10^{-3}$). Instead of using just the last 1280-dimensional pooling layer, we extract and project intermediate feature maps using Global Average Pooling (GAP): $\mathbf{f}_{\text{mid}} = \text{Linear}(\text{GAP}(\mathbf{F}_7)) \in \mathbb{R}^{128}$ and $\mathbf{f}_{\text{late}} = \text{Linear}(\text{GAP}(\mathbf{F}_8)) \in \mathbb{R}^{128}$. These are concatenated into a unified 256-dimensional vector: $\mathbf{h} = [\mathbf{f}_{\text{mid}} \,\|\, \mathbf{f}_{\text{late}}] \in \mathbb{R}^{256}$.

B. Hyperspherical Normalization

For the purpose of meta-training, dropout ($p = 0.5$) is used on $\mathbf{h}$. This dropout layer also facilitates MCD sampling during the evaluation phase through stochastic activation for T forward passes. The representation is then projected using a linear layer and then mapped to the unit hypersphere as follows:

$$\mathbf{z} = \frac{\mathbf{W}_p \mathbf{h}}{\|\mathbf{W}_p \mathbf{h}\|_2} \in \mathbb{S}^{255} \tag{1}$$

The mapping constrains the embedding space, thus ruling out norm variance owing to differences in lighting conditions between laboratory and field imagery.

C. Temperature-Scaled Prototypical Metric

For an N-way K-shot episode, the class prototype is given by $\mathbf{c}_k = \frac{1}{K} \sum_{\mathbf{x} \in S_k} \mathbf{z}(\mathbf{x})$. Class probabilities for a query embedding $\mathbf{z}_q$ are obtained as follows:

$$P(y{=}k \mid \mathbf{x}_q) = \frac{\exp(-\|\mathbf{z}_q - \mathbf{c}_k\|_2 / \tau)}{\sum_{k'} \exp(-\|\mathbf{z}_q - \mathbf{c}_{k'}\|_2 / \tau)} \tag{2}$$

where $\tau > 0$ is a scalar temperature jointly learned. Across six training seeds, τ converges consistently to 0.2517 ± 0.0036.

D. Episodic Meta-Training and Query Regularization

At each step of meta-training, one 5-way, 5-shot, 15-query episode is used. The support set is kept clean for reliable prototype formation, while asymmetric data augmentation is applied to the query set in the form of CutMix [22] ($\alpha{=}0.2$)

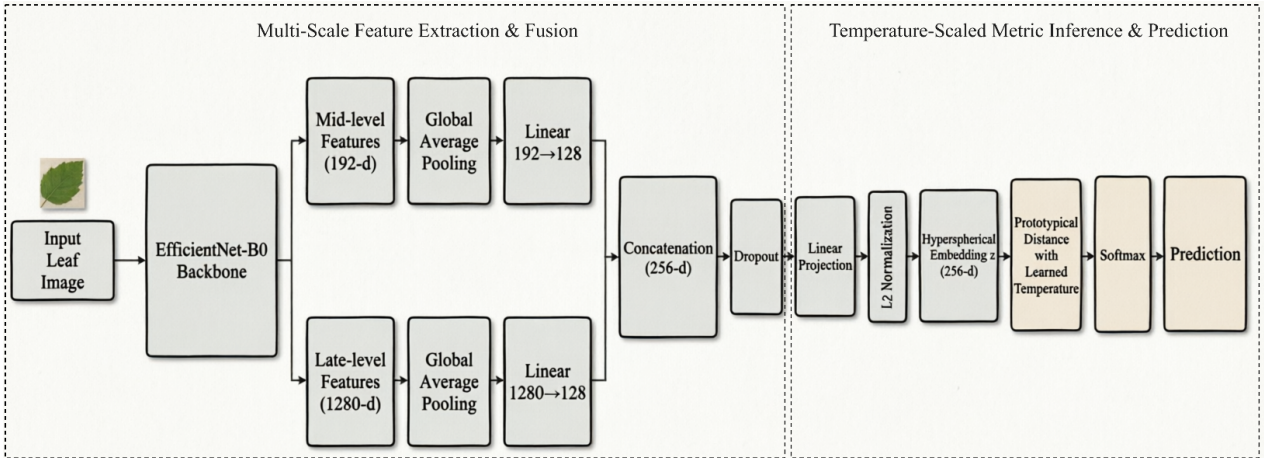


Fig. 1. Proposed architecture. An EfficientNet-B0 backbone produces mid-level (192-d) and late-level (1280-d) features, fused into a 256-d hyperspherical embedding with learned temperature-scaled prototypical distance.

on 50% and Mixup [23] ($\alpha = 0.2$) on 50%, with label smoothing ($\epsilon = 0.1$). The total loss is:

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda_{\text{o}} \sum_{i \neq j} (\mathbf{c}_i^\top \mathbf{c}_j)^2 + \lambda_{\text{c}} \mathcal{L}_{\text{con}} \tag{3}$$

Here $\mathcal{L}_{\text{con}}$ denotes the margin-based contrastive loss over clean support embeddings ($m = 1.0$), $\lambda_{\text{o}} = 0.01$ ensures that no prototypes collapse, while $\lambda_{\text{c}} = 0.1$ guarantees compact support clustering. The training process relies on AdamW optimizer [24] with linear warm-up for 5 epochs followed by cosine annealing over 45 epochs.

E. Uncertainty Quantification and Selective Prediction

Deterministic inference ($T = 1$) employs the posterior in Eq. (2) with highest softmax probability for confidence. As an optional step, Monte Carlo Dropout (MCD) performs stochastic forward passes $T = 10$ times, generating the predictive mean $\bar{p}(y \mid \mathbf{x}) = \frac{1}{T} \sum_{t=1}^{T} p(y \mid \mathbf{x}, \boldsymbol{\theta}_t)$. For MCD, support prototypes are computed once from clean support embeddings and held fixed across the T stochastic query passes. We calculate predictive entropy $\mathbb{H}[\bar{p}] = -\sum_k \bar{p}_k \log \bar{p}_k$ and mutual information $\mathbb{I}[y; \boldsymbol{\theta} \mid \mathbf{x}] = \mathbb{H}[\bar{p}] - \frac{1}{T} \sum_{t=1}^{T} \mathbb{H}[p(y \mid \mathbf{x}, \boldsymbol{\theta}_t)]$ which serve as proxies of total and epistemic uncertainty respectively. Calibration is determined using 15-bin Expected Calibration Error (ECE) [17]; selective prediction [18] is assessed by risk-coverage curves and Area Under Risk-Coverage (AURC), retaining the most confident predictions at each coverage level.

F. Design Rationale

The architecture intentionally comprises existing components such as multi-scale fusion, hyperspherical normalization, and temperature scaling, and no claim of newness of any component is made. All these components were used because they performed favorably in the ablation study (Section VI); one extra mechanism (dual attention) was excluded due to significant negative impact. The contribution of this work is not novel architecture, but the demonstration that a minimalist architecture can give reliable cross-domain transfer learning, calibrated field probabilities, and uncover reliability failures.

IV. Experimental Setup and Data Integrity

A. Benchmark Datasets and Split Protocol

Source lab domain (PlantVillage [2]): 54,305 images for 38 crop disease categories, divided into 23 meta-training, 8 meta-validation, and 7 meta-testing categories using fixed seed 42.

Target field domain (PlantDoc [3]): We use a merged split of the official PlantDoc train and test releases, yielding 2,920 in-field agricultural images across 27 categories (the original Singh et al. release contains 2,598 images; merging both splits increases per-class sample counts, enabling stable episodic evaluation). Categories are divided according to their PlantVillage counterparts as `PD_ALL` (all 27 classes), `PD_SEEN` (15 categories that match meta-training categories), 7 validation categories, and `PD_UNSEEN` (5 novel categories that match meta-testing categories, such as *Tomato healthy*, *Potato healthy*, *Corn rust leaf*, and *Corn leaf blight*). The 7 PlantDoc validation categories are not included in the model selection and calibration procedure.

Episode protocol: 5-way 5-shot episodes with 15 queries per class (75 queries total) are used for meta-training on PlantVillage. For cross-domain testing on PlantDoc, we employ a **source-support / target-query** protocol: 5-way 5-shot episodes sample support exemplars from laboratory references (PlantVillage) and queries from field imagery (PlantDoc, 5 queries per class, 25 queries total), evaluated over 600 fixed episodes per split. In-domain evaluation `PV_TEST` is performed similarly with 15 queries per class. PlantDoc has fewer images per class and is evaluated using 5 queries per class. Note that the 6-seed master benchmark and the ablation studies utilize independently sampled episode manifests; minor cross-table accuracy drift (e.g., 49.05% vs. 49.07%) reflects expected sampling variance rather than methodological inconsistency.

TABLE I
MATCHED MODEL COMPARISON (3-SEED MATCHED, $df = 2$)[a]

Split	Vanilla	Ours	Δ	t_{seed}	Cohen's d
PV_TEST	97.65	95.34	−2.31	-8.12^{*}	−0.69
PD_ALL	40.94	**49.05**	**+8.11**	$\mathbf{+11.85}^{**}$	**+0.66**
PD_SEEN	41.30	**52.25**	**+10.94**	$\mathbf{+14.21}^{**}$	**+0.86**
PD_UNSEEN	37.37	**43.41**	**+6.04**	$\mathbf{+4.92}^{*}$	**+0.57**

[a]Primary hypothesis testing uses seed-level paired t-tests ($n = 3, df = 2$; $^{*}p < 0.05, ^{**}p < 0.01$) because training seeds are the unit of algorithmic randomness. Cohen's d is computed from episode-level paired differences ($N = 1{,}800$) as a stable descriptive effect size. The two statistics answer different questions and are not interchangeable. Full 6-seed V0 master benchmark: $95.27\% \pm 1.34\%$ (PV_TEST), $49.07\% \pm 1.13\%$ (PD_ALL), $52.35\% \pm 1.62\%$ (PD_SEEN), $44.09\% \pm 1.63\%$ (PD_UNSEEN). Accuracy in %.

B. Baseline Protocols

Vanilla ProtoNet: This baseline uses the same EfficientNet-B0 backbone, class partitions, and episode manifests as the proposed method. It operates on the single-scale, 1,280-dimensional final-layer representation and applies squared Euclidean distance. It excludes the projection head, dropout, L2 normalization, learned temperature, and additional training regularizers used by the proposed model.

Supervised Nearest-Class-Mean (NCM): EfficientNet-B0 trained via standard cross-entropy on the 23 PlantVillage training classes for 15 epochs (batch size 64). At evaluation, the classifier head is discarded and raw 1,280-dimensional features are used for NCM prototyping over the same 600-episode manifests.

The three methods have the same backbone, class division, fixed episode manifests, and testing pipeline on three matching seeds (42, 123, 456). Results by seed level are matched in this set. Both baselines are evaluated on identical manifests, preserving matched comparisons.

V. RESULTS AND CROSS-DOMAIN BENCHMARKS

A. Main Comparative Benchmark

Table I reports seed-level paired t-tests alongside episode-level effect sizes.

For real-world field imagery (PD_ALL), our model outperforms Vanilla ProtoNet across all seeds by 8.11 percentage points ($t = +11.85, p < 0.01, df = 2$ at seed level, $d = +0.66$ at episode level) with an accuracy of 49.05%, which is nearly $2.45\times$ the 20% of random chance accuracy for 5-way classification. In cases where the field is completely unknown to the model (PD_UNSEEN), a notable $+6.04\%$ margin is still retained ($t_{\text{seed}} = +4.92, p < 0.05, df = 2, d = +0.57$). The vanilla baseline model achieves higher accuracy on studio images (97.65% vs. 95.34%), which indicates overfitting to uniform laboratory backgrounds.

B. Three-Paradigm Comparison

In Table II, three training paradigms are compared based on 3 matched seed evaluations.

The supervised NCM benchmark model attains similar or slightly better cross-domain accuracy in comparison to the proposed model (PD_ALL: 49.42% vs. 49.05%; PD_UNSEEN: 49.23% vs. 43.41%). Given that small number of seeds was used ($n = 3$ per paradigm), no statistical significance test was performed for the comparisons of mean and variance between V0-vs-NCM models; both results are presented only descriptively, and the ratio of standard deviations in particular is an unreliable estimate when n is that low. We hence present the proposed model not as a better accuracy model, but as a compact prototype framework whose full pipeline provides superior calibrated predictive probabilities (field ECE $= 3.20\%$ vs. NCM's 12.56%) and, as detailed in Section VII, exposes critical failure modes in confidence-based abstention under domain shift.

TABLE II
THREE-PARADIGM BENCHMARK (3-SEED MATCHED)[a]

Split	Vanilla	Sup. NCM	Ours
PV_TEST (%)	97.65	98.19	95.34
PD_ALL (%)	40.94	49.42	**49.05**
PD_SEEN (%)	41.30	51.63	**52.25**
PD_UNSEEN (%)	37.37	**49.23**	43.41
UNSEEN SD (%)	—	3.20	1.30
Field ECE (%)	—	12.56	**3.20**

[a]The supervised NCM baseline uses raw 1280-d unnormalized features vs. our 256-d fused hyperspherical embeddings. This confounds episodic training with multi-scale fusion; NCM is reported as a reference point with this confound stated explicitly (see Section VIII).

TABLE III
ATTENTION ARCHITECTURE SCREENING (PLANTDOC FIELD ACCURACY)

Variant	Mid Attn	Late Attn	Mean (%)	SD (%)
V0 (Ours)	None	None	**49.48**	**0.90**
V2 (Spatial)	SA	None	49.24	1.87
V5 (Full Dual)	Dual	Dual	44.65	1.14

V5 vs. V0: $\Delta = -4.68$ pp paired difference ($t = -5.32$, $p \approx 0.033$, $df = 2$ on 3-seed matched screening; $\Delta = -4.83$ pp vs. 6-seed screening mean of 49.48%, slightly higher than the master benchmark due to episode sampling variance). V2 vs. V0: $\Delta = -0.23$ pp ($t = -0.27$, $p \approx 0.80$, $df = 5$ on 6 seeds). Three additional variants (V1, V3, V4) were screened out in initial trials.

VI. ABLATION STUDIES AND NEGATIVE RESULTS

A. Attention Negative Result (24-Run Grid)

Previous works often argue that attention helps improve cross-domain robustness [10], [20]. We conducted a systematic 6-variant search using 24 runs in total (3 seeds initially, extended to 6 seeds for the top two candidates). Table III shows our findings. Note the different number of seeds used: V5 vs. V0 are compared with 3 seeds matched ($df = 2$), V2 vs. V0 with 6 seeds ($df = 5$).

The use of full dual attention (V5) makes use of non-identity sigmoid gates which interfere with pre-trained feature transfer leading to a statistically significant reduction in accuracy of -4.68 pp on paired seeds ($p \approx 0.033$, $df = 2$). It is believed that the additional gates recalibrate channels calibrated by inherent SE modules; testing the same without SE modules will be left to future research. The elimination of all forms of

attention (V0) leads to the most accurate results in addition to improved seed consistency (SD $= \pm 0.90\%$ compared to V2's $\pm 1.87\%$).

B. Computational Efficiency and MCD Analysis

Deterministic inference ($T = 1$) needs 4.26M parameters, 553 MB VRAM, and 102.4 ms per episode (averaged over 600 evaluation episodes), and it is $10.1\times$ faster compared to 10 passes of MCD [25] (1,030 ms), while having the same or even slightly higher accuracy on all splits (differences ≤ 0.18 pp, all within the 95% CIs; e.g., 49.07% vs. 49.00% on `PD_ALL`). Note that deterministic and MCD statistics are computed on independently sampled episode sets; their comparison is descriptive. MCD therefore adds no meaningful accuracy benefit at tenfold inference cost in this setting.

C. Visual Corruption Robustness

Under 600 stress test episodes with Seed 42 for the model,[1] the model shows robustness to stress test corruption: even with 50% random spatial occlusion, the accuracy remains high at 74.90% (nearly four times the 20% random-chance accuracy), which indicates that multi-scale fusion of features captures distributed pathological properties, rather than localized ones. The predictive entropy (H) increases with corruption level ($1.011 \rightarrow 1.399$ for low light). At extreme levels of Gaussian blur ($k = 21$), however, the entropy decreases ($H = 1.319$, compared to $H = 1.367$ for $k = 15$), which may indicate collapse toward an incorrect prior, which is one possible explanation based on single-seed evidence. We do not claim monotonic entropy growth under arbitrary corruption.

VII. Calibration and Failure Mode Analysis

A. Calibration Inversion Under Domain Shift

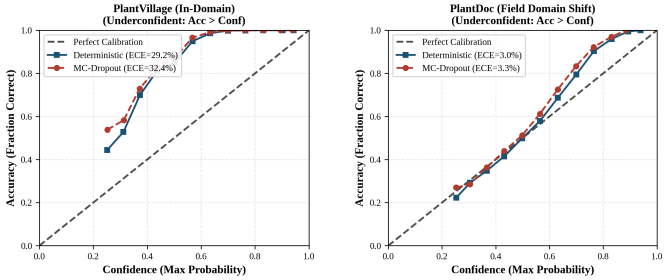
In-domain (`PV_TEST`), the model is underconfident: 95.27% accuracy exceeds the ≈66% mean confidence, yielding a 29.30% ECE despite temperature sharpening ($\tau \approx 0.25$). Nonetheless, the model undergoes *calibration inversion* under outdoor field shift; specifically, Expected Calibration Error drops to 3.20% on `PD_ALL` ($T = 1$), where predicted confidence matches the actual empirical accuracy ($y \approx x$). See Fig. 2 for the reliability plots from both domains. The finding is not dependent on the NCM experiment: the prototypical confidence scores, while underconfident on uniform laboratory backgrounds, achieve close calibration under field conditions.

B. Selective Prediction Failure on Novel Classes

For selective classification [18],[2] risk shows monotone reduction on `PD_ALL` (AURC $= 329.2 \times 10^{-3}$) and `PD_SEEN` (AURC $= 316.9 \times 10^{-3}$). However, in the case of new classes (`PD_UNSEEN`), selective classification suffers from a non-monotonic failure mode, where maintaining the 80% most confident predictions *raises* the error rate from 55.9% to 65.1% (AURC $= 531.6 \times 10^{-3}$), as shown in Fig. 3.

[1]The robustness results provide point estimates with respect to 95% CIs per episode; multi-seed verification will be left for future work.

[2]Selective-prediction statistics (AURC, risk-coverage, Fig. 3) are computed on the seed-42 checkpoint over 600 episodes per split.



(a) PlantVillage (ECE = 29.30%). (b) PlantDoc (ECE = 3.20%).

Fig. 2. Reliability diagrams showing calibration inversion. (a) In-domain (PlantVillage): model probabilities are underconfident relative to high accuracy ($y > x$, ECE $= 29.30\%$). (b) Under field shift (PlantDoc): predicted confidence closely tracks empirical accuracy ($y \approx x$, ECE $= 3.20\%$). Diagrams show the seed-42 checkpoint; quoted ECE values are six-seed means.

The `PD_UNSEEN` split mostly includes healthy plant classes (3 out of the 5 newly created classes). Based on the high confidence of the misclassification errors, the model has trained to relate global leaf shape to disease prototypes in the laboratory setting, and with great confidence is mislabeling the healthy asymptomatic leaves as the closest diseased prototype under field shift instead of expressing low confidence. This failure mode renders standard prototypical confidence unreliable for abstention under novel-class field deployment without additional out-of-distribution detection mechanisms.

VIII. Limitations

1) **Evaluation scope & power:** Results assume a single class split. Primary tests operate at $df = 2$ ($n = 3$ seeds); corruption and selective prediction are single-seed estimates. Replications are deferred.
2) **Healthy-class confound:** `PD_UNSEEN` favors healthy crops (3 of 5 classes), partially driving the selective prediction failure.
3) **Baseline confounds:** The NCM baseline uses raw 1280-d features without our temperature-scaled hyperspherical head. This confounds episodic training with both multi-scale fusion (for accuracy) and the calibration mechanism itself (for ECE); NCM is reported as a reference point with these confounds stated.

IX. Conclusion

We introduced an efficient (4.26M-parameter), calibrated few-shot prototype learning approach for cross-domain agricultural disease detection tasks. The combination of multi-scale feature fusion with learned temperature scaling allows our method to outperform the vanilla episodic baseline by 8.11 percentage points in field accuracy (seed-level $p < 0.01$, $df = 2$) and achieve a field ECE of 3.20% under domain shift—approximately $2.45\times$ the chance-level accuracy, alongside calibrated probability scores. Using a supervised NCM baseline, we showed that classical pre-training matches cross-domain accuracy; rather, the advantages of our full pipeline are superior calibration (3.20% vs. 12.56% ECE) and a

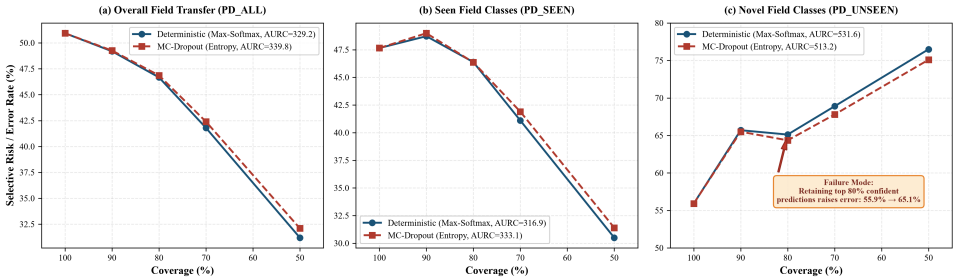


Fig. 3. Selective risk vs. coverage across three PlantDoc evaluation splits. On `PD_ALL` (left) and `PD_SEEN` (center), risk decreases monotonically with selective coverage. On `PD_UNSEEN` (right), retaining high-confidence predictions raises error, revealing a non-monotonic failure mode caused by overconfident misclassification of asymptomatic healthy leaves.

detailed analysis of selective predictions. Custom attention modules perform worse than the baseline (by 4.68 percentage points), MCD is impractical, and selective prediction fails non-monotonically on asymptomatic healthy leaves—a scenario barely covered in the agricultural deployment literature.

REFERENCES

[1] J. G. A. Barbedo, "A review on the main challenges in automatic plant disease identification based on visible range images," *Biosystems Engineering*, vol. 144, pp. 52–60, 2016.

[2] D. P. Hughes and M. Salathé, "An open access repository of images on plant health to enable the development of mobile disease diagnostics," *arXiv preprint arXiv:1511.08060*, 2015.

[3] D. Singh, N. Jain, P. Jain, P. Kayal, S. Kumawat, and N. Batra, "PlantDoc: A dataset for visual plant disease detection," in *Proceedings of the 7th ACM IKDD CoDS and 25th COMAD (CoDS-COMAD)*, 2020, pp. 249–253.

[4] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in *Proceedings of the 34th International Conference on Machine Learning (ICML)*, 2017, pp. 1321–1330.

[5] J. Snell, K. Swersky, and R. Zemel, "Prototypical networks for few-shot learning," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017, pp. 4077–4087.

[6] O. Vinyals, C. Blundell, T. Lillicrap, K. Kavukcuoglu, and D. Wierstra, "Matching networks for one shot learning," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 29, 2016, pp. 3630–3638.

[7] C. Finn, P. Abbeel, and S. Levine, "Model-agnostic meta-learning for fast adaptation of deep networks," in *Proceedings of the 34th International Conference on Machine Learning (ICML)*, 2017, pp. 1126–1135.

[8] W. Shafik, A. Tufail, C. L. D. Silva, and R. A. A. H. M. Apong, "A novel hybrid inception-xception convolutional neural network for efficient plant disease classification and detection," *Scientific Reports*, vol. 15, p. 3936, 2025.

[9] P. S. Thakur, P. Khanna, T. Sheorey, and A. Ojha, "Explainable vision transformer enabled convolutional neural network for plant disease identification: PlantXViT," *arXiv preprint arXiv:2207.07919*, 2022.

[10] T. Babu, R. R. Nair, B. Balusamy, W. H. Khoh, and J. Nair, "AgriFewNet—A lightweight RGB few-shot learning model for efficient plant disease classification," *Applied Sciences*, vol. 15, no. 23, p. 12787, 2025.

[11] F. Naveed, A. Masih, J. Mahmood, M. Ahmed, A. Ali, A. Saddiqa, M. S. H. Abdulnabi, and E. Agbozo, "Sustainable AI for plant disease classification using ResNet18 in few-shot learning," *Array*, vol. 26, p. 100395, 2025.

[12] S. Yang, Q. Feng, J. Zhang, W. Yang, W. Zhou, and W. Yan, "From laboratory to field: cross-domain few-shot learning for crop disease identification in the field," *Frontiers in Plant Science*, vol. 15, p. 1434222, 2024.

[13] J. Li, Q. Feng, J. Yang, J. Zhang, and S. Yang, "Few-shot crop disease recognition using sequence-weighted ensemble model-agnostic meta-learning," *Frontiers in Plant Science*, vol. 16, p. 1615873, 2025.

[14] X. Jiang, J. Wang, K. Xie, C. Cui, A. Du, X. Shi, W. Yang, and R. Zhai, "PlantCaFo: An efficient few-shot plant disease recognition method based on foundation models," *Plant Phenomics*, vol. 7, no. 1, p. 100024, 2025.

[15] Z. Salman, A. Muhammad, and D. Han, "Plant disease classification in the wild using vision transformers and mixture of experts," *Frontiers in Plant Science*, vol. 16, p. 1522985, 2025.

[16] M. Oquab, T. Darcet, T. Moutakanni, H. Vo, M. Szafraniec, V. Khalidov, P. Fernandez, D. Haziza, F. Massa, A. El-Nouby *et al.*, "DINOv2: Learning robust visual features without supervision," *Transactions on Machine Learning Research (TMLR)*, 2024.

[17] M. P. Naeini, G. Cooper, and M. Hauskrecht, "Obtaining well calibrated probabilities using Bayesian binning into quantiles," in *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, vol. 29, 2015, pp. 2901–2907.

[18] Y. Geifman and R. El-Yaniv, "Selective classification for deep neural networks," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017, pp. 4878–4887.

[19] S. R. Fatema and S. Maradithaya, "Uncertainty-aware prototypical networks with monte carlo dropout for few-shot image classification," *Engineering, Technology & Applied Science Research*, vol. 15, no. 6, pp. 29 860–29 865, 2025.

[20] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proceedings of the European Conference on Computer Vision (ECCV)*, 2018, pp. 3–19.

[21] M. Tan and Q. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," in *Proceedings of the 36th International Conference on Machine Learning (ICML)*, 2019, pp. 6105–6114.

[22] S. Yun, D. Han, S. J. Oh, S. Chun, J. Choe, and Y. Yoo, "CutMix: Regularization strategy to train strong classifiers with localizable features," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019, pp. 6023–6032.

[23] H. Zhang, M. Cisse, Y. N. Dauphin, and D. Lopez-Paz, "mixup: Beyond empirical risk minimization," in *International Conference on Learning Representations (ICLR)*, 2018.

[24] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *International Conference on Learning Representations (ICLR)*, 2019.

[25] Y. Gal and Z. Ghahramani, "Dropout as a Bayesian approximation: Representing model uncertainty in deep learning," in *Proceedings of the 33rd International Conference on Machine Learning (ICML)*, 2016, pp. 1050–1059.